JIANWEN LYU

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EDUCATION

Northwestern University Evanston, IL
Master of Science in Computer Science Sept 2024 – Sept 2026

- **GPA**: 3.92/4.0.
- **Selected courses**: Machine Learning, Intro to Computer Graphics, Natural & Artificial Vision, Deep Learning.

University of Nottingham Nottingham, UK
Bachelor of Science with Honors, Computer Science with Artificial Intelligence Sept 2022 – Jul 2024

- **GPA**: First Class Honors, 76/100.
- **Selected courses**: Languages and Computation, Machine Learning, Artificial Intelligence Methods.

University of Nottingham Ningbo China Ningbo, China
Bachelor of Science, Computer Science with Artificial Intelligence Sept 2020 – Jun 2022

- **GPA**: First Class Honors, 80/100.
- **Awards**: UNNC University Scholarship (Jun 2021, Jun 2022).

PUBLICATIONS

Yunfei Ge, Anbang Liu, Qineng Wang, Johnalbert Garnica, **Jianwen Lyu**, Zihan Wang, et al. (2026). MINDTOPO: Can Foundation Models Reason in Topological Space? *NeurIPS 2026* NEmo Workshop. Accepted, Oral Presentation.
Qineng Wang, Wenlong Huang, Yu Zhou, Hang Yin, Tianwei Bao, **Jianwen Lyu**, Weiyu Liu, Ruohan Zhang, Jiajun Wu, Li Fei-Fei, Manling Li.(2026). ENACT: Evaluating Embodied Cognition with World Modeling of Egocentric Interaction. *International Conference on Learning Representations*, *NeurIPS 2025 Workshop on Learning with Limited Labeled Data (LAW)*, Spotlight Presentation.
Jianwen. Lyu, N. Lane. (2024). Hyperparameter Search and Performance Assessment of CNNs for Image Classification. Accepted at the *International Conference on Signal Processing and Machine Learning*.

RESEARCH AND PROJECT EXPERIENCES

RoboARC: Coding Agents as Policy Generators for Constrained Robotic Manipulation
Project Lead, supervised by Dr. Manling Li, Northwestern University Jun 2026 - Present

- Lead RoboARC, a physics-based benchmark evaluating coding agents as policy generators for long-horizon manipulation under sustained physical constraints.
- Defined manipulation capabilities and a task taxonomy spanning contact, threading, interlocking, and deformable objects; designed most tasks and developed most simulation infrastructure.
- Built controlled environments and reproducible evaluation protocols to assess constraint satisfaction and support fine-grained trajectory analysis and systematic diagnosis of manipulation failures.
- Extended TopoGym, a topology-sensitive VLA evaluation benchmark, into RoboARC to study agent-generated control policies. Preparing a first-author manuscript on RoboARC.

SentenceSSM: Exploring Sentence as an Input Granularity for Selective State-Space Models
Master's Thesis, supervised by Dr. Manling Li, Northwestern University Jun 2026

- Proposed SentenceSSM, an end-to-end architecture that keeps a Mamba2 backbone unchanged and replaces its token-level input with whole-sentence SONAR embeddings to investigate whether token-level inputs provide sufficient semantic context for Mamba's content-based selection mechanism.
- Designed and implemented the 780M-parameter architecture and training pipeline, including sentence-space input/output adapters, trajectory packing, cross-entropy training through the SONAR decoder, 14.4B-token FineWeb pretraining on 4×A100 GPUs, and a budget-matched token-level Mamba2 baseline.
- Built a three-stage evaluation covering generation quality, zero-shot language benchmarks, and a mechanistic Δt gate-selectivity probe.
- Demonstrated that SentenceSSM's selective gate consistently concentrated on a priori important sentences across 100 RULER variable-tracking trajectories (mean rank 4.76 of ~710; Wilcoxon $p \approx 4\times10^{-18}$; Cliff's $\delta = 1.0$), providing a clean mechanistic observation consistent with the granularity-mismatch hypothesis.

Integrating MDP Formulations with Concept-level Reasoning Models in Language Models
Research Assistant, MLL Lab, Northwestern University Apr 2025 – Present

- Developed a unified framework integrating Markov Decision Process (MDP) formulations with concept-level reasoning models to formalize decision-making and abstraction in language-based reasoning systems.
- Designed model architectures and research methodology to combine MDP-based decision structure with concept-level reasoning.
- Conducted theoretical analysis and empirical evaluation to study how the integration supports structured reasoning and improves the formal interpretability of language-based reasoning systems.
- Preliminary experiments provide initial evidence for the effectiveness of the framework on structured reasoning tasks, offering a promising direction for unifying sequential decision-making and abstract concept-level reasoning in machine learning systems.

The Evaluation of Embodied Reasoning in Vision-Language Model
Research Assistant, supervised by Dr.Manling Li, Northwestern University Sept 2025

- Investigated how vision-language models (VLMs) understand egocentric interactions and real-world state changes to uncover structural, semantic, and handedness-related failure modes.
- Actively drove the design of the ENACT benchmark and conducted ablation experiments and data collection; developed a component-level error analysis framework parsing scene-graph and state-change differences into structured signatures, enabling systematic categorization of structural and semantic errors.
- Performed statistical, qualitative, and handedness-asymmetry analyses to examine left–right hand confusion and agent-interaction failures across multiple VLMs and human predictions.
- Revealed that current VLMs exhibit substantially higher error rates than humans on embodied reasoning tasks, with omission and hallucination accounting for over 40% of structural errors, semantic failures reflecting difficulty in correctly tracking left–right hand use during egocentric interactions.

The Evaluation of Topological Reasoning and Planning for Multimodal Large Language Models (MLLMs)
Research Assistant, supervised by Dr.Manling Li, Northwestern University Sept 2025

- Developed the MINDTOPO benchmark for evaluating topological reasoning and planning abilities of MLLMs across topological primitives such as continuity, separation, order, enclosure, and knots.
- Designed a 2D maze-based topological environment, deployed and maintained evaluation servers for multiple API-accessible foundation models; participated in human annotation of topological data.
- Revealed a major gap between visual topological recognition and actionable reasoning in foundation models, with the best model achieving only 54.13% accuracy compared to a 97.40% human ceiling.

The Design of An Interactive Data-managing Interface in JupyterLab
Principal Investigator, supervised by Dr. Kai Xu, University of Nottingham Mar 2024 – Jun 2024

- Designed and developed an interactive interface as a JupyterLab-based extension for AI-related data management and operations to improve the interactivity, usability, and interpretability.
- Investigated the interface design choices to improve user interaction and tool interpretability.
- Conducted a systematic report on the tool's design, implementation, and integration within the JupyterLab ecosystem, with attention to both technical feasibility and research usability.

Investigation of CNN Overfitting regarding Regularization and Training
Remote Research Assistant, supervised by Dr. Nicholas Lane, University of Cambridge Jul 2023 – Aug 2023

- Designed a lightweight convolutional neural network (CNN) to investigate overfitting dynamics and improve model generalization through regularization and training strategies on the Dogs vs. Cats benchmark.
- Conducted comparative experiments on training strategies and hyperparameter optimization to analyze their effects on overfitting behavior and generalization performance; systematically evaluated model configurations.
- Achieved 85% classification accuracy and identified training configurations, demonstrating the practical value of regularization and hyperparameter tuning in CNN-based image classification.

Analyzing the Effects of Data Preprocessing and Feature Variation on Supervised Learning Models
Participant, supervised by Dr. Nicolas Lane, University of Oxford Aug 2022 – Sept 2022

- Examined the effects of data preprocessing and feature variation on model behavior and evaluation outcomes to explore how supervised learning models perform under different feature configurations.
- Applied Python and scikit-learn to preprocess real-world datasets and evaluate supervised learning models under varying feature configurations.
- Conducted comparative analyses of model performance and parameter sensitivity to assess how feature design and parameter choices influence predictive behavior.

Internship Experiences

Chinese Academy of Sciences, Institute of Automation, *Intern*, Beijing China Jul 2023 – Oct 2023

- Investigated learning stability, convergence behavior, and policy optimization in deep reinforcement learning by implementing a Deep Q-Network (DQN) with convolutional neural architectures and experience replay on the Cart-Pole benchmark in OpenAI Gym.
- Evaluated convergence and learning stability, producing analytical reports with performance curves and tuning results; implemented and evaluated Proximal Policy Optimization (PPO) in the SMAC multi-agent benchmark, documenting policy behavior and comparative performance across experimental settings.
- Achieved stable policy convergence for DQN on the Cart-Pole benchmark in OpenAI Gym and generated reproducible analyses of architecture design, update dynamics, and hyperparameter sensitivity.

Shandong Domor Intelligence Technology Co., Ltd., *Front-end Developer*, Jinan, China Jun 2022 – Aug 2022

- Developed interactive web modules in JavaScript for the Dust Monitoring Platform, focusing on interactive data display, user-facing functionality, and integration of real-time environmental information into the web interface.
- Enabled more effective real-time visualization and analysis of dust-monitoring data, contributing to a more usable and responsive platform for environmental data observation and operational decision support.

Additional Information

- **Academic Services**: Reviewer, FMEA Workshop (CVPR 2026); LLA Workshop (COLM 2026).
- **Programming**: Python, JavaScript, C# (for Unity-based game development).
- **Tools/Frameworks**: PyTorch, scikit-learn, OpenAI Gym, SMAC (StarCraft II Multi-Agent Challenge), accelerate.
- **Languages**: Chinese (Native), English (Proficient).